

Courses List

Name: Jingyi Le

Undergraduate: King's College London, UK (2022-2025)

Postgraduate: University of Cambridge, UK (2025-2026)

These courses are listed in chronological order.

I. Undergraduate: 1st Year

1. Calculus I

Lecturer: Prof. Eugene Shargorodsky

Textbook: Spivak, Michael, *Calculus*

Description: Injective, surjective and bijective functions; trigonometric functions; hyperbolic functions; inverse functions; the logarithm and exponential functions; limits; differentiation from first principles; integration from first principles; series; Taylor's theorem and l'Hopital's.

Grade: 94

2. Calculus II

Lecturer: Dr. Calum Spicer

Textbook:

Description: Surface sketching, partial derivatives, multiple integrals, geometry of curves, vector fields, geometry of surfaces, maxima and minima, generalised derivatives, Stokes' Theorem, the Divergence Theorem.

Grade: 94

3. Linear Algebra I

Lecturer: Dr. Nazar Miheisi

Textbook:

Description: Complex numbers. Vector spaces: row and column vectors (real and complex), linear combinations, abstract vector spaces, bases and dimension of vector spaces. Linear maps: definitions and examples, matrix of a linear map, composition, invertibility and isomorphisms. Solving systems of linear equations using linear algebra. Determinants: definition and properties. Eigenvectors and eigenvalues: definitions, diagonalisation of matrices, canonical forms of 2x2 matrices.

Grade: 100

4. Linear Algebra II

Lecturer: Prof. Konstanze Rietsch; Dr. Asuka Kumon

Textbook: Treil, Sergei, *Linear Algebra Done Wrong*

Description: Inner product spaces: inner products for real and complex vector spaces, orthogonal and orthonormal spaces, projections. Special classes of matrices, including unitary and orthogonal matrices. Spectral theory, including the Cayley-Hamilton theorem, eigenspaces and multiplicities, Jordan normal form. Quadratic forms: definitions, diagonalisation and geometric applications.

Grade: 93

5. Sequence and Series

Lecturer: Prof. David Burns; Dr. Asuka Kumon

Textbook:

Description: An introduction to sets and functions: defining sets, subsets, intersections and unions; injections, surjections, bijections. Compositions and inverses of functions. An introduction to mathematical logic and proof: logical operations, implication, equivalence, quantifiers, converse and contrapositive; proof by induction and contradiction, examples of proofs. Then these ideas will be applied in the context of the real numbers to make rigorous arguments with sequences and series and develop the notions of convergence and limits.

Grade: 92

6. Introduction to Algebra

Lecturer: Dr. Peter Jossen; Dr. Beth Romano

Textbook: Dummit, David Steven, *Abstract Algebra*

Description: The integers: well-ordering principle, division and the Euclidean algorithm. Prime numbers and the fundamental theorem of arithmetic. Binary operations, equivalence relations and quotient sets. Modular arithmetic. Groups: definition and examples, abelian and cyclic groups, orders of group elements, cosets and Lagrange's theorem. Rings: definitions and examples, unit groups. The Chinese remainder theorem. Polynomials.

Grade: 86

7. Introduction to Dynamical Systems

Lecturer: Prof Gérard Watts; Dr Eleni Kontou

Textbook:

Description: Differential equations; first-order dynamical systems, autonomous systems, phase flow and fixed points; second-order dynamical systems, phase flow, classification of fixed points; kinematics of particle motion, Newton's laws; conservation of energy, conservative forces, motion on a straight line; Hamiltonian systems; elements of Hamiltonian mechanics.

Grade: 95

8. Probability and Statistics I

Lecturer: Dr. Dominic Yeo; Dr. Claudia Neves

Textbook: Dekking, Michel, *A modern introduction to probability and statistics*

Description: Elementary combinatorial analysis, Definition of probability, Unions and intersections - Statistical Independence, Conditional probability, Bayes's theorem, Random variables (discrete and continuous), Probability distributions (Binomial, Poisson, Normal, Gamma), Covariance and correlation, Independent variable and the law of large numbers, Central limit theorem, Sampling distributions (Normal, Students-t), Estimators, Confidence intervals, Hypothesis testing.

Grade: 85

II. Undergraduate: 2nd Year

1. Applied Differential Equations

Lecturer: Prof. Giuseppe Tinaglia

Textbook: Nagle, R. Kent; Saff, E. B.; Snider, Arthur David, *undamentals of differential equations and boundary value problems*

Description: Basics of ODEs. Fourier series. PDEs, separation of variables.

Grade: 94

2. Complex Analysis

Lecturer: Prof. Eugene Shargorodsky

Textbook: Priestley, H. A., *Introduction to complex analysis*

Description: Holomorphic functions, Laurent series, residue calculus, conformal maps.

Grade: 73

3. Real Analysis

Lecturer: Dr. Natalie Evans

Textbook:

Description: Rigorous treatment of continuity, differentiation and Riemann integral for functions of one variable.

Grade: 94

4. Geometry of surface

Lecturer: Dr. David Sheard

Textbook: Pressley, Andrew, *Elementary differential geometry*

Description: Definition of a curve, arc length, curvature and torsion of a curve, Frenet-Serret equations. Definition of a surface patch, first and second fundamental forms, isometries, conformal maps, area, Gaussian curvature, mean curvature, principal curvatures, Gauss

map, geodesics, Theorema Egregium.

Grade: 93

5. Introduction to Number Theory

Lecturer: Dr. Steve Lester

Textbook: Stein, William, *Elementary Number Theory: Primes, Congruences, and Secrets: A Computational Approach*

Description: Review of divisibility, prime numbers and congruences. Residue class rings, Euler's function, primitive roots. Quadratic residues and quadratic reciprocity law. Irrational and transcendental numbers. Sums of squares. Some Diophantine equations. If time permits we will also learn about the 'AKS' algorithm for primality testing.

Grade: 94

6. Classical Dynamics

Lecturer: Prof. Christopher Herzog

Textbook: Gregory, R. Douglas, *Classical Mechanics*

Description: This course aims to give an introduction to the basic concepts and tools used to study all dynamical systems, from pendulums to falling cats, to the planets, stars and galaxies. The course will cover Newtonian mechanics in three dimensions, vector calculus, inertial frames, linear and angular momentum, multi-particle systems, work, energy, conserved quantities, conservative forces, potentials, central forces, configuration space, Lagrangian systems, the principle of least action, Euler-Lagrange equations, generalised coordinates, constrained systems, Kepler's laws, Hamiltonian dynamics, Poisson brackets, symmetries, Noether's theorem, Liouville's theorem and the Poincare recurrence theorem. Newtonian, Lagrangian, Hamiltonian mechanics.

Grade: 90

7. Groups and Symmetries

Lecturer: Dr. Paul P. Cook

Textbook: Humphreys, J. F., *A course in group theory*

Description: To be able to construct and classify finite groups of low order; to be able to use the homomorphism theorem, the direct product and the semi-direct product to construct groups; to be able to work with the classical matrix groups; to understand the structure of $SU(2)$ and $SO(3)$; to study the isomorphisms of Euclidean spaces.

Grade: 87

8. Discrete Mathematics

Lecturer: Prof. Payman Kassaei; Dr. Lassina Dembele

Textbook:

Description: Elementary properties of Integers. Functions and their behaviour. Introduction to Recursion. Algorithms and complexity. Graphs including Euler's Theorem, shortest path algorithm and vertex colouring. Trees - applications include problem solving and spanning trees. Directed Graphs including networks. Codes and Cyphers - with Hamming codes and RSA.

Grade: 90

III. Undergraduate: 3rd Year

1. Probability and Statistics II

Lecturer: Dr. Guillaume Conchon-Kerjan; Dr. Luca Galimberti

Textbook:

Description: Bivariate probability, continuous densities, generating functions. The exponential densities, including normal, t -, X^2 and F . Simple parametric and nonparametric tests. Further topics include the consistency, efficiency and sufficiency of estimates, maximum likelihood estimation; the central limit theorem, Chebyshev's inequality, the Neyman-Pearson lemma and the likelihood ratio test; regression, analysis of variance.

Grade: 92

2. Fourier Analysis (Graduate level)

Lecturer: Prof. Alexander Pushnitski

Textbook: Stein, Elias M.; Shakarchi, Rami, *Fourier analysis: An introduction*

Description: Series expansions. Definition of Fourier series. Related expansions. Bessel's inequality. Pointwise and uniform convergence of Fourier series. Periodic solutions of differential equations. The vibrating string. Convolution equations. Mean square convergence. Schwartz space S . Fourier transform in S . Inverse Fourier transform. Parseval's formula. Solutions of differential equations with constant coefficients.

Grade: 97

3. Galois Theory (Graduate level)

Lecturer: Dr. Netan Dogra; Dr Andrea Dotto

Textbook: Stewart, Ian, *Galois theory*

Description: Review of the basic theory of rings, polynomials and fields; Eisenstein's Criterion; first properties of finite extensions of fields and their degrees; algebraicity and transcendence; field embeddings and automorphisms; normal extensions; separable extensions; the Galois Correspondence; examples of practical calculation; soluble groups and extensions; (in)solubility of polynomial equations by radical expressions. Further topics may include finite fields, constructability by straightedge and compass, etc. as time allows.

Grade: 65

4. Fundamentals of Probability Theory

Lecturer: Dr. Dominic Yeo

Textbook:

Description: Countability, measure spaces, σ -algebras, π -systems and uniqueness of extension. Construction of Lebesgue measure on $\mathbb{R}$ (proof non-examinable), Independence. The Borel-Cantelli lemmas, measurable functions and random variables, independence of random variables. Notions of probabilistic convergence. Construction of integral and expectation. Integration and limits. Density functions. Product measure and Fubini's theorem. Laws of large numbers. Characteristic functions and weak convergence, Gaussian random variables. The central limit theorem. Conditional probability and expectation.

Grade: 88

5. Statistical Inference

Lecturer: Dr. Marina Riabiz

Textbook: Casella, George; Berger, Roger L., *Statistical inference*

Description: Estimation: unbiasedness, consistency, Cramér–Rao bound. Fisher information. Sufficiency. Robustness. Methods of estimation: maximum likelihood estimators, asymptotic properties. Confidence intervals. Method of least squares. Generalised method of moments. Prior and posterior distributions. Bayesian estimators and Bayesian intervals. Hypothesis testing: Likelihood ratio. Neyman-Pearson lemma. Power function. Two-sided tests. Composite hypothesis. Bayesian testing.

Grade: 74

6. Rings and Modules

Lecturer: Dr. Peter Jossen

Textbook:

Description: Basic concepts of ring theory: subrings, ideals, quotient, product, matrix and polynomial rings; factorisation in integral (euclidean, principal ideal) domains. Basic concepts of module theory: submodules, quotient modules, direct sums, homomorphisms, finitely generated, cyclic, free and torsion modules, annihilator ideals. Matrices and finitely generated modules over a principal ideal domain: Equivalence of matrices, structure theory of modules, applications to abelian groups and to vector spaces with a linear transformation.

Grade: 87

7. Representation Theory of Finite Groups

Lecturer: Dr. Samuel Johnston

Textbook: Ledermann, Walter, *Introduction to group characters*

Description: The basic definitions and standard properties of linear representations of finite groups over the complex numbers (in particular Schur's lemma and Maschke's theorem). The relation between representations and characters, the orthogonality relations and other

fundamental properties of characters and character tables. Application of the above results to performing explicit calculations for groups of small order.

Grade: 82

8. Numerical and Computational Methods with Python

Lecturer: Dr. Nikolas Nüsken and Prof. Benjamin Doyon

Textbook: Kiusalaas, Jaan, *Numerical methods in engineering with Python 3*

Description: Solution of non-linear equations. Approximation of functions by polynomials. Numerical differentiation and integration. Numerical solution of ordinary differential equations, and systems of linear equations. Rates of convergence, and errors. The algorithms developed will be implemented in Python. A short introduction to the modern industrial strength package Wolfram Mathematica will be given.

Grade: 99

IV. Postgraduate: Part III Year

I'm currently taking Part III, here's courses I attend, the exams will be held in June 2026. All these courses are graduate level.

1. Local Fields

Lecturer: Prof. Tom Fisher

Textbook: J.W.S. Cassels, *Local fields*

Description: Topics to be covered include: absolute values on fields, valuations, complete fields and their extensions, inverse limits, Hensel's lemma and ramification theory. If time permits, then possible further topics include: local class field theory (statements only), the Hilbert norm residue symbol, and the Hasse-Minkowski theorem.

Grade:

2. Analytic Number Theory

Lecturer: Dr. Joni Teravainen

Textbook: T. Apostol, *Introduction to Analytic Number Theory*

Description: In this course we will cover results on the distribution of prime numbers, in particular the celebrated prime number theorem. This is done by studying analytic properties of the Riemann zeta function and connections between the location of its zeros and results on prime numbers. We will also cover results from the theory of multiplicative functions, such as mean value results for these. This is a topic that has seen a lot of recent activity.

Grade:

3. Diophantine Analysis

Lecturer: Prof. Peter Varju

Textbook: E. Bombieri and W. Gubler, *Heights in Diophantine geometry*

Description: This course will discuss two classical methods in Diophantine analysis. The first one can be traced back to Thue, who proved the following result in Diophantine approximation. The second method that will be discussed in the course goes back to the work of Baker. Both methods have been utilized by many authors for a wide range of applications in number theory and beyond. In the course, we will discuss both methods, and we will sample from their applications.

Grade:

4. Elliptic Curves

Lecturer: Dr. Adam Morgan

Textbook: J.H. Silverman, *The Arithmetic of Elliptic Curves*

Description: Elliptic curves are the first non-trivial curves, and it is a remarkable fact that they have continuously been at the centre stage of mathematical research for centuries. This will be an introductory course on the arithmetic of elliptic curves, concentrating on the study of the group of rational points

Grade:

5. Commutative Algebra

Lecturer: Dr. Nicholas Williams

Textbook: M. F. Atiyah and I. G. Macdonald., *Introduction to commutative algebra*

Description: This course will provide an introduction to the theory of commutative rings and modules over these rings. It should be viewed as a foundational course for Algebraic Geometry and Algebraic Number Theory.

Grade:

6. Algebraic Methods in Combinatorics

Lecturer: Dr. Mansur Boase

Textbook: Alon, N, *Combinatorics*

Description: This course will cover a selection of the many beautiful applications of algebra in combinatorics, ranging from the 18th to the 21st century. The purpose is to introduce and motivate some of these methods and offer a foundation for what is a very active area of research.

Grade:

7. Algebraic Geometry

Lecturer: Prof. Mark Gross

Textbook: R. Hartshorne, *Algebraic Geometry*

Description: The course will be an introduction to modern algebraic geometry. The first part of the course will develop the basic theory of sheaves, and use this to give a treatment of schemes, morphisms between them, and fibre products. We will then explore morphisms the basic properties of morphisms, including separatedness and properness. After this, we turn to the study of quasicoherent sheaves, with a particular focus on line bundles, the Picard group, and other proximate notions. The final few lectures in the course will give a rapid introduction to sheaf cohomology “for the working mathematician”. The course will develop the basic toolkit used by practicing algebraic geometers, and will also be appropriate for anyone with an interest in acquiring a working knowledge of algebraic geometry, including those in nearby fields such as differential geometry, topology, and number theory.

Grade:

8. Essay: Sarnak’s Conjecture on Möbius Disjointness

Essay setter: Dr. Joni Teravainen

Textbook:

Description: A sequence $(a(n))_{n \in \mathbb{N}}$ taking values ± 1 is said to be of low complexity if the number of different length k sign patterns $(a(n+1), \dots, a(n+k))$ that appear in the sequence grows slower than exponentially in k . This notion of low complexity can also be extended to complex-valued sequences using dynamical language. Sarnak’s conjecture states that the Möbius function should be orthogonal to all low complexity sequences $(a(n))_{n \in \mathbb{N}}$.

Despite a lot of progress in recent years, this conjecture remains open.

Grade: